

Homework 2

1. (a) Let $\mathcal{B} = \bigcup_{p \in \Gamma} \mathcal{B}_p$. For each $p \in \Gamma$ there exists $B \in \mathcal{B}$ with $p \in B$ by construction. Let $B_1, B_2 \in \mathcal{B}$ and $q \in B_1 \cap B_2$.

If q has $y > 0$, then B_1, B_2 are usual open discs in the upper half-plane, hence there exists $\varepsilon > 0$ such that $B_d(q, \varepsilon) \subset B_1 \cap B_2$. Since $\varepsilon < y_q$, we have $B_d(q, \varepsilon) \in \mathcal{B}_q$, so the basis condition holds.

If $q = (x, 0)$, then any basis element at q has the form $\{q\} \cup A$, where A is an open disc tangent at q . The intersection of two such sets contains $\{q\}$ together with a smaller tangent disc at q , which is again a basis element in $\mathcal{B}_q$.

Thus $\mathcal{B}$ satisfies the basis axioms and determines a topology T on Γ , called the Moore plane.

- (b) For $p = (x, y)$ with $y > 0$, the sets $B_d(p, 1/n)$, $n \in \mathbb{N}$, with $1/n < y$, form a countable local base. For $p = (x, 0)$, let A_n be the open disc of radius $1/n$ tangent at p ; then $\{p\} \cup A_n$, $n \in \mathbb{N}$, is a countable local base. Thus (Γ, T) is first countable. Let $D = \{(x, y) \in \Gamma : x, y \in \mathbb{Q}, y > 0\}$. If $U \in T$ contains a point with $y > 0$, it contains a usual open disc in the upper half-plane, therefore a point of D . If U contains $(x, 0)$, it contains $\{(x, 0)\} \cup A$ with A a tangent disc, and A contains a point of D . Thus D is dense and (Γ, T) is separable.
- (c) No. If (Γ, T) were second countable, then every subspace would be second countable. By (d), the x -axis with the subspace topology is discrete and uncountable, thus not second countable. Therefore (Γ, T) is not second countable.
- (d) Let $X = \{(x, 0) : x \in \mathbb{R}\}$. For each $p \in X$, the set $\{p\}$ equals $(\{p\} \cup A) \cap X$ for any tangent disc A at p , therefore is open in the subspace topology. Thus every subset of X , being a union of singletons, is open. We can then conclude that the subspace topology on the x -axis is discrete.

2. (a) Since $A \subset A \cup B$ and $B \subset A \cup B$, monotonicity of closure gives

$$\text{Cl}(A) \cup \text{Cl}(B) \subset \text{Cl}(A \cup B).$$

Conversely, if $x \notin \text{Cl}(A) \cup \text{Cl}(B)$, then there exist open sets U, V with $x \in U$, $x \in V$, $U \cap A = \emptyset$, and $V \cap B = \emptyset$. Then $U \cap V$ is open, contains x , and is disjoint from $A \cup B$, hence $x \notin \text{Cl}(A \cup B)$. Thus

$$\text{Cl}(A \cup B) = \text{Cl}(A) \cup \text{Cl}(B).$$

- (b) By definition, $\partial A = \text{Cl}(A) \cap \text{Cl}(X \setminus A)$. If $x \in A$, then every neighborhood of x meets A , so $x \in \text{Cl}(A)$. Hence $A \subset \text{Cl}(A)$. If $x \in \text{Cl}(A)$, then either $x \in A$ or $x \notin A$, in which case every neighborhood of x meets both A and $X \setminus A$, so $x \in \partial A$. Thus

$$\text{Cl}(A) = A \cup \partial A.$$

Homework 2

MA 564
Feb 17

- (c) If $x \in \text{Int}(A)$, then some open U satisfies $x \in U \subset A$, hence $x \notin \partial A$. Thus $\text{Int}(A) \subset A \setminus \partial A$. Conversely, if $x \in A \setminus \partial A$, then $x \notin \text{Cl}(X \setminus A)$, so there exists open $U \ni x$ with $U \cap (X \setminus A) = \emptyset$, hence $U \subset A$ and $x \in \text{Int}(A)$. Therefore

$$\text{Int}(A) = A \setminus \partial A.$$

Moreover, $x \notin \text{Int}(A)$ iff every neighborhood of x meets $X \setminus A$, i.e. $x \in \text{Cl}(X \setminus A)$. Thus

$$\text{Int}(A) = X \setminus \text{Cl}(X \setminus A).$$

- (d) From (b) and (c),

$$\partial A = \text{Cl}(A) \setminus \text{Int}(A).$$

- (e) By (c), $\text{Int}(A) \cap \partial A = \emptyset$. Also $\text{Int}(X \setminus A) = X \setminus \text{Cl}(A)$, so it is disjoint from both $\text{Cl}(A)$ and $\partial A \subset \text{Cl}(A)$. Finally, for any $x \in X$, either $x \in \text{Int}(A)$, or $x \in \text{Int}(X \setminus A)$, or every neighborhood of x meets both A and $X \setminus A$, in which case $x \in \partial A$. Thus

$$X = \text{Int}(A) \cup \partial A \cup \text{Int}(X \setminus A),$$

and the union is disjoint.

3. (a) For $x \in X$, every neighborhood of x meets $\{x\}$, hence $x \in \text{Cl}(\{x\})$ and $x \leq_T x$; thus $\leq_T$ is reflexive.

If $x \leq_T y$ and $y \leq_T z$, then $x \in \text{Cl}(\{y\})$ and $y \in \text{Cl}(\{z\})$. Let U be an open set with $x \in U$. Since $x \in \text{Cl}(\{y\})$, we have $y \in U$. Since $y \in \text{Cl}(\{z\})$, we have $z \in U$. Thus every neighborhood of x contains z , so $x \in \text{Cl}(\{z\})$ and $x \leq_T z$. Therefore $\leq_T$ is transitive, and therefore a pre-order.

- (b) Let U be open and suppose $x \in U$ with $x \leq_T y$. If $y \notin U$, then $X \setminus U$ is closed and contains y , hence contains $\text{Cl}(\{y\})$, so $x \notin U$, a contradiction. Thus $y \in U$, and U is an upper set.

If F is closed and $x \in F$ with $y \leq_T x$, then $y \in \text{Cl}(\{x\}) \subset F$, so F is a lower set.

- (c) If $y \in \uparrow x$, then $x \in \text{Cl}(\{y\})$, so every neighborhood of x contains y . Thus y lies in the intersection of all neighborhoods of x . Conversely, if y lies in every neighborhood of x , then every open set containing x contains y , thus $x \in \text{Cl}(\{y\})$ and $y \in \uparrow x$. Therefore

$$\uparrow x = \bigcap \{U : U \text{ is a neighborhood of } x\}.$$

By definition,

$$\downarrow x = \{y \in X : y \leq_T x\} = \text{Cl}(\{x\}).$$

- (d) $\downarrow x = \text{Cl}(\{x\})$ is always closed. In general, $\uparrow x$ need not be open, since it is an arbitrary intersection of open sets.

Homework 2

(e) Let $X = \{0, 1\}$ and $\mathcal{T} = \{\emptyset, \{1\}, \{0, 1\}\}$. Closed sets are $\emptyset, \{0\}, \{0, 1\}$. Thus

$$\text{Cl}(\{0\}) = \{0\}, \quad \text{Cl}(\{1\}) = \{0, 1\}.$$

Therefore $x \leq_T y$ iff $x \in \text{Cl}(\{y\})$, so

$$0 \leq_T 0, \quad 1 \leq_T 1, \quad 0 \leq_T 1,$$

and no other relations hold. Thus $0 \leq_T 1$, and $1 \not\leq_T 0$.

4. Let $Y = [-1, 1] \subset \mathbb{R}$ with the subspace topology.

Set	Open in Y	Open in $\mathbb{R}$
$A = \{x : \frac{1}{2} < x < 1\}$	Yes	Yes
$B = \{x : \frac{1}{2} < x \leq 1\}$	Yes	No
$C = \{x : \frac{1}{2} \leq x < 1\}$	No	No
$D = \{x : \frac{1}{2} \leq x \leq 1\}$	No	No
$E = \{0 < x < 1 \text{ and } \frac{1}{x} \notin \mathbb{N}\}$	Yes	Yes

5. Let $\mathbb{R}_0$ denote $\mathbb{R}$ with the discrete topology and $\mathbb{R}_1$ denote $\mathbb{R}$ with the standard topology. On $\mathbb{R} \times \mathbb{R}$ consider the dictionary order:

$$(x_1, y_1) < (x_2, y_2) \quad \text{iff} \quad x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

If $a = (x, y)$, $b = (x, z)$ with $y < z$, then

$$(a, b) = \{x\} \times (y, z).$$

If $a = (x, y)$ and $b = (x', z)$ with $x < x'$, then

$$(a, b) = (\{x\} \times (y, \infty)) \cup ((x, x') \times \mathbb{R}) \cup (\{x'\} \times (-\infty, z)).$$

Since $\mathbb{R}_0$ is discrete, every subset of $\mathbb{R}$ is open; thus sets of the form $\{x\} \times U$, with U open in $\mathbb{R}$, form a basis for the product topology $\mathbb{R}_0 \times \mathbb{R}_1$. The above descriptions show that every basic open interval in the dictionary order is open in $\mathbb{R}_0 \times \mathbb{R}_1$, and conversely every set $\{x\} \times (y, z)$ is a basic open interval in the dictionary order. Thus the two topologies coincide.

In $\mathbb{R}_0 \times \mathbb{R}_1$, each vertical line $\{x\} \times \mathbb{R}$ is open. In the standard topology on $\mathbb{R}^2$, no vertical line is open. Therefore the dictionary (product) topology is strictly finer than the standard topology on $\mathbb{R}^2$.

6. (a) Let $(X, \mathcal{T})$ be second countable with countable basis $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$. Let $\mathcal{U}$ be an open cover of X . For each B_n such that $B_n \subset U$ for some $U \in \mathcal{U}$, choose one such U and denote it by U_n . Then $\{U_n : n \in \mathbb{N}\}$ is countable.

If $x \in X$, choose $U \in \mathcal{U}$ with $x \in U$. Since $\mathcal{B}$ is a basis, there exists B_n with $x \in B_n \subset U$. Hence $x \in U_n$. Thus $\{U_n : n \in \mathbb{N}\}$ is a countable subcover, and X is Lindelöf.

Homework 2

MA 564

Feb 17

- (b) **(b)** Every second countable space is Lindelöf by (a).

Conversely, let (X, d) be a Lindelöf metric space. For each $n \in \mathbb{N}$, the family $\{B(x, 1/n) : x \in X\}$ is an open cover of X . By Lindelöf, there exists a countable set $D_n \subset X$ such that

$$X = \bigcup_{x \in D_n} B(x, 1/n).$$

Let $D = \bigcup_{n \in \mathbb{N}} D_n$. Then D is countable. The collection

$$\{B(x, 1/n) : x \in D, n \in \mathbb{N}\}$$

is countable and forms a basis for X . Thus X is second countable.

- (c) Let $\mathbb{R}_\ell$ denote $\mathbb{R}$ with the lower limit topology. Consider the set

$$D = \{(x, -x) : x \in \mathbb{R}\} \subset \mathbb{R}_\ell \times \mathbb{R}_\ell.$$

For each $x \in \mathbb{R}$, set

$$U_x = [x, x+1) \times [-x, -x+1).$$

Each U_x is open and $(x, -x) \in U_x$, so $\{U_x : x \in \mathbb{R}\}$ covers D .

If $x \neq y$, then $(x, -x) \notin U_y$; hence each U_x contains exactly one point of D . Thus any subcover of D must contain all U_x , so no countable subcollection covers D .

If $\mathbb{R}_\ell \times \mathbb{R}_\ell$ were Lindelöf, then its subspace D would be Lindelöf. This contradiction shows that $\mathbb{R}_\ell \times \mathbb{R}_\ell$ is not Lindelöf.

7. Let $(\mathbb{R}, \mathcal{T})$ be $\mathbb{R}$ with the cofinite topology (closed sets are finite or all of $\mathbb{R}$). Let $E = \{2, 4, 6, \dots\}$.

- (a) In the cofinite topology, every nonempty open set is dense. So, it becomes clear that every infinite set is dense in $\mathbb{R}$.

$$\text{Cl}(E) = \mathbb{R}, \quad L(E) = \mathbb{R} \setminus E, \quad \partial E = \text{Cl}(E) \setminus \text{Int}(E) = \mathbb{R} \setminus \emptyset = \mathbb{R}.$$

- (b) A countable set $D \subset \mathbb{R}$ is dense since any nonempty open set is cofinite and intersects D . Thus $(\mathbb{R}, \mathcal{T})$ is separable.
- (c) Suppose $(\mathbb{R}, \mathcal{T})$ with the cofinite topology is pseudometrizable. Every pseudometrizable space is first countable: if d is a pseudometric, then

$$\{B_d(x, 1/n) : n \in \mathbb{N}\}$$

is a countable local base at x .

However, in the cofinite topology on an uncountable set, no point has a countable local base. Indeed, if $\{U_n\}_{n \in \mathbb{N}}$ were a local base at x , then each U_n is cofinite, so

$$\bigcap_{n \in \mathbb{N}} U_n$$

Homework 2

MA 564

Feb 17

is cofinite and therefore contains some $y \neq x$. But then every neighborhood of x would contain y , contradicting that $\mathbb{R} \setminus \{y\}$ is an open neighborhood of x not containing y .

Thus $(\mathbb{R}, \mathcal{T})$ is not first countable and therefore not pseudometrizable.

8. (a) True. Let $\mathcal{B}$ be a basis for T' . If f is continuous, then $f^{-1}(B)$ is open for each $B \in \mathcal{B}$ because B is open.

Conversely, suppose $f^{-1}(B)$ is open for every $B \in \mathcal{B}$. Let $U \in T'$. Then U is a union of basis elements: $U = \bigcup_{i \in I} B_i$, $B_i \in \mathcal{B}$. Then

$$f^{-1}(U) = f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i),$$

which is open in X as a union of open sets. Thus f is continuous.

- (b) Let $\mathcal{S}$ be a sub-basis for T' . If f is continuous, clearly $f^{-1}(S)$ is open for each $S \in \mathcal{S}$.

Conversely, if $f^{-1}(S)$ is open for all $S \in \mathcal{S}$, then $f^{-1}(U)$ is open for any finite intersection of sub-basis elements. However, a general open set in T' is a union of finite intersections of sub-basis elements. Thus $f^{-1}(U)$ is a union of open sets and is open, so the converse holds as well.

9. Let X be an infinite Hausdorff space. We construct inductively a sequence $\{x_n\}_{n \in \mathbb{N}}$ of distinct points and open sets U_n such that

$$x_n \in U_n \quad \text{and} \quad U_n \cap \{x_k : k \neq n\} = \emptyset.$$

Choose $x_1 \in X$. Suppose $x_1, \dots, x_n$ have been chosen. Since X is infinite, pick $x_{n+1} \in X \setminus \{x_1, \dots, x_n\}$.

For each $k \leq n$, by the Hausdorff property there exist disjoint open sets V_k and W_k such that $x_{n+1} \in V_k$ and $x_k \in W_k$. Let

$$U_{n+1} = \bigcap_{k=1}^n V_k.$$

Then U_{n+1} is open, contains x_{n+1} , and is disjoint from $\{x_1, \dots, x_n\}$.

For each $k \leq n$, replace U_k by $U_k \cap W_k$; this remains open, contains x_k , and is disjoint from x_{n+1} . Thus we obtain pairwise disjoint open sets $U_1, \dots, U_{n+1}$ with $x_k \in U_k$.

Proceeding inductively yields distinct points $\{x_n\}_{n \in \mathbb{N}}$ and pairwise disjoint open sets U_n with $x_n \in U_n$.

Let $Y = \{x_n : n \in \mathbb{N}\}$. For each n ,

$$U_n \cap Y = \{x_n\},$$

so $\{x_n\}$ is open in the subspace topology on Y . Thus every subset of Y is open, and Y is an infinite discrete subspace.

Homework 2

MA 564

Feb 17

10. Let $f : [0, 2\pi) \rightarrow S^1$ be given by $f(\theta) = (\cos \theta, \sin \theta)$, where both spaces carry the subspace topology from $\mathbb{R}$ and $\mathbb{R}^2$, respectively.

The map $\theta \mapsto (\cos \theta, \sin \theta)$ is continuous from $\mathbb{R}$ to $\mathbb{R}^2$. Since $[0, 2\pi)$ and S^1 have the subspace topologies, the restriction f is continuous.

Consider the open set $U = [0, \varepsilon)$ in $[0, 2\pi)$, where $\varepsilon > 0$. Then U is open in the subspace topology.

Its image $f(U)$ is an arc of S^1 containing $(1, 0)$ but only from one side. Any open neighborhood of $(1, 0)$ in S^1 contains points corresponding to angles near both 0 and 2π , whereas $f(U)$ contains no points corresponding to angles near 2π . Thus $f(U)$ is not open in S^1 .

Therefore f is continuous but not open.